Highlights

Calibrated gating of corrupted input tokens in a pretrained text-image masked diffusion model: a preregistered measurement on UniDisc

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- **[RESULT: see highlights.txt; written after the results are frozen]**

Calibrated gating of corrupted input tokens in a pretrained text-image masked diffusion model: a preregistered measurement on UniDisc

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Abstract

A unified masked diffusion model treats every observed token as certain, although the tokens of a real input can be damaged by compression, noise, splicing or a mismatched caption. This paper asks whether a pretrained model can find damaged image tokens itself, and whether withholding them changes a caption-verification decision. Each image token is scored by its masked likelihood inside the image with the caption withheld. A threshold is set by conformal risk control on clean image-caption pairs, so that the expected proportion of clean tokens flagged is at most a chosen level. Flagged tokens are removed from an additive image-to-caption evidence statistic. The measurement is preregistered, runs on one pretrained 1.4B model with no network finetuning, and compares the gate with no gate, with random masks of the same size and shape, with cheap non-generative detectors and with a pixel denoiser. **[RESULT: headroom outcome (which corruptions leave room for a gate), realised false-alarm proportion with interval, primary paired contrast per corruption with interval and Holm-adjusted p-value, mechanism and detection controls. Use only the wording allowed by the preregistration, section 12; one sentence on scope: one model, one dataset.]**

Keywords: Masked discrete diffusion, Conformal risk control, Input reliability, False-alarm control, Vision-language tokens, Fault detection

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1. Introduction

Input-token gating asks which of the observed tokens given to a model should be withheld before the model reasons from them. The question arises for any model that reads a sequence of discrete tokens as evidence, and it is sharp for masked discrete diffusion models. A masked diffusion model is trained to predict any masked token from any visible subset of the sequence, and the mask token means "unknown" [1, 2, 3]. Every token that is visible is therefore treated as certain. This is acceptable for a clean training sequence. It is a weakness when the input comes from a pipeline that compresses, adds noise to, crops, splices or mislabels the data, because a single wrong visible token is used with the same confidence as a correct one. Unified models extend the setting to two or more modalities in one token sequence [4, 5, 6, 7]. In such a model a damaged image token can change what the model concludes about a caption, and the reverse.

Why this is a fusion problem.. Each image token and each caption token is a source that reports a value, and the model combines them. The paired decision studied here, whether a caption describes an image, sums per-token log-likelihood ratios over the image tokens. That is the log-likelihood-ratio rule used to fuse local decisions in distributed detection, in which each source is weighted by its reliability before a global threshold is applied [8, 9]. The literature on multisensor fusion treats imperfect and spurious sensor data as a first-order difficulty [10], and decision fusion with unreliable or falsified sources has its own review [11]. Classical practice answers this difficulty in two steps. It first checks each source for a fault and isolates the faulty ones before fusing, a task treated under fault detection and isolation [12, 13]. It sets the detection threshold from clean data so that the false-alarm rate stays at a chosen value, the idea behind constant false-alarm rate (CFAR) detection [14, 15]. In this paper the image tokens are the sources, the masked denoiser is the model of what a healthy source looks like, and the gate is the fault detector that runs before the evidence is summed. This is a framing of the measurement, not a claim of a new fusion rule: the evidence statistic is the standard additive likelihood-ratio sum, and the contribution lies in the gate and in how it is calibrated and measured.

What exists and what is missing.. Many recent methods remask tokens in a masked diffusion model. Most act on tokens that the sampler itself produced: they remask low-confidence commitments, learn a quality score, or perturb

the context to find brittle tokens [16, 17, 18, 19, 20, 21, 22]. In multimodal masked diffusion models the same idea has been applied to the model's own generations, with cross-modal confidence as the trigger [23, 24]. Work on corrupted *input* tokens is thinner. Trust but Verify [25] detects corrupted conditioning tokens in a text-only masked diffusion model, re-noises them to the mask state and refills them; it needs a finetune, uses a hand-set threshold, and has no second modality. Conformal risk control has been used with masking-derived scores to flag erroneous caption words given an image, with a CLIP-based score and a threshold calibrated on labelled foil data [26]. The authors of UniDisc show repair of corrupted image-caption pairs by random masking and best-of-n re-infilling in one qualitative figure [4]. Section 2 gives the precise differences. Three things are missing from this picture. No measurement separates a gate that finds damaged tokens from a mask that merely has the same shape. No threshold for input tokens carries a stated bound on the false-alarm proportion on clean inputs. And no comparison shows whether withholding flagged tokens helps a cross-modal decision more than cheap non-generative detectors or a pixel denoiser do.

Contribution, stated as a measurement.. This paper does not claim to be the first to detect or remask corrupted input tokens, the first to remask in a unified model, the first risk-controlled threshold for vision-language token errors, or the first masked-likelihood or null-normalised generative score. The claim is the following, with every outcome clause a preregistered hypothesis. On a pretrained unified text-image masked diffusion model, with no network finetuning, each input token is scored by its masked likelihood in its own modality with the other modality withheld. The flag threshold is set by conformal risk control on clean pairs, so that the expected false-alarm proportion on clean inputs is at most a chosen level α. The paper then measures whether withholding or re-infilling the flagged tokens preserves cross-modal verification under image-token corruption, against no gate, equal-size random gates, cheap detectors and a pixel denoiser. The contributions are:

- *A gate and its calibration* (Section 3). The gate statistic, the evidence-separation rule that keeps cross-modal terms out of the gate, the conformal threshold with the exact guarantee it gives and the guarantees it does not give, and seven algorithms that build from one masked-score primitive to the full verification procedure.

- *A preregistered protocol* (Section 4). One primary hypothesis family with a fixed correction, controls that separate detection from mask geometry, from texture and from denoising, a blind headroom rule that removes corruptions that leave nothing to repair, and a stated reading of null results.

- *Measurements on one model* (Section 5). **[RESULT: one sentence per confirmed finding, with the number and interval from the frozen run files and the preregistered wording; findings that the data do not support are stated as such.]**

- *An exploratory feasibility result* (Section 4). A pilot on 300 clean image-caption pairs shows the pair-level evidence signal that the gate and the endpoint rely on.

The scope is one pretrained 1.4B model, one image-caption dataset, three realistic image corruptions and two sanity corruptions. Re-infilling, caption corruption and two-way fusion are described in the method but are outside the preregistered main run; Section 4 states which parts were run.

[AUTHOR: Figure 1 (motivation, page 1-2 per IF anatomy rule 31): one measured clean versus corrupted example with the per-token evidence map, from a committed script; to be made after the results are frozen. Do not draw it with a generative image tool.]

Roadmap.. Section 2 places the work against the closest methods. Section 3 defines the scores, the calibrated gate, the evidence statistic and the algorithms. Section 4 gives the data, folds, corruptions, arms, baselines, endpoints, tests and hardware, with the exploratory pilot. Section 5 reports the results, Section 6 discusses them with the limitations, and Section 7 concludes.

2. Related work

The related work is organised along four axes: which tokens are treated as suspect, how a token is scored, how the threshold is set, and what is done with a flagged token. Table 1 lists the closest works on these axes.

Masked diffusion and unified models.. Absorbing-state discrete diffusion [1, 27] was simplified into masked diffusion language models [2] and scaled [28]. A masked diffusion denoiser models the conditional distribution of a clean token given the unmasked context [3], which is the property the gate score uses.

Masked image-token generation [29] and unified text-image models [4, 5, 6, 7] put several modalities in one sequence. UniDisc, the backbone used here, is trained to mask all tokens of a modality with some probability, so that a modality likelihood with the other modality withheld exists [4]. The appendix of that paper demonstrates repair of overlaid and mismatched image-caption pairs by random masking of both modalities, re-unmasking and best-of-n selection by the model's own likelihood, shown as examples without a metric or a per-token score. That recipe is the closest prior procedure for re-infilling and is treated as a baseline in Section 4.

Remasking and self-correction of generated tokens.. Training-free remasking acts on the sampler's own commitments [16, 20, 21]. Learned variants add a per-token quality score or a correction head [17, 18, 19]. For masked image generation, a learned critic decides which sampled image tokens to resample [22]. In multimodal masked diffusion models, self-correction re-evaluates committed tokens with the token remasked and remasks the lowest-scoring ones [23]. A coupled sampler for concurrent text and image generation retracts commitments when cross-modal evidence turns against them [24]. All of these act on tokens that the model produced. The tokens studied here are given evidence whose provenance is untrusted, which raises a question that does not arise for self-correction: whether a suspect token should be withheld (left masked) or imputed (re-infilled). The loop of detecting, remasking and refilling is not claimed as new, and the cross-modal remasking trigger belongs to the works above.

Corrupted input and conditioning tokens.. Trust but Verify [25] is the nearest work. It studies corrupted context tokens in a masked diffusion language model with 170M parameters, finetuned with substituted context tokens. Its suspicion score is the model's probability of the observed token at its unmasked position, its threshold is annealed by hand within an edit budget derived from an assumed corruption rate, and flagged tokens are re-noised to the mask state and refilled. The paper reports a false-edit rate on clean tokens, and it reports that corruptions that are plausible under the left context are only partly detected even with the finetune. The present work differs in five ways: no finetuning of the network, a score computed after masking the token, with the other modality withheld, a second modality, a threshold with a stated bound on the clean false-alarm proportion, and an outcome measured on a cross-modal decision rather than on infilling quality.

Its observation that detection of plausible corruptions is hard is treated here as an expected difficulty for JPEG and noise, with a stated falsifier (Section 4). That paper also suggests that bidirectional consistency checking is needed for plausible corruptions; the masked score used here is one way to make that check. Because the finetune is what makes the observed-position probability informative in that paper, the baseline named "TbV-score (non-finetuned)" below reproduces its score but not its method.

In vision-language models, uncertainty-guided dropout decoding masks visual tokens that are uncertain under a logit-lens projection, with hand-set drop probabilities and without a calibrated threshold [30]. A self-purification defence masks visual tokens of high saliency at inference to remove backdoor behaviour [31]. Neither evaluates detection of corrupted tokens against known labels.

Masked-likelihood scores, null-normalised scores and calibrated thresholds.. Scoring a token by its masked likelihood has a long history in text: the pseudo-log-likelihood [32], its variant for multi-piece words [33], and unsupervised spelling detection from the probability of the original token under a masked language model [34]. A generative model can be used for image-text matching by normalising the conditional error with the unconditional one [35], a null-normalised score that the pointwise-mutual-information statistics of this paper resemble. None of these is claimed as new. Conformal prediction supplies distribution-free guarantees under exchangeability [36, 37], conformal p-values give false-discovery control for outlier detection [38], and conformal risk control bounds the expected value of a monotone loss [39]. Gomes et al. [26] apply conformal risk control with masking-derived CLIPScore distributions to flag erroneous caption words. Their guarantee is a high-probability upper bound (Hoeffding and Bentkus) calibrated on foil-labelled data, with a CLIP score that changes when words or patches are masked. The calibration here uses clean pairs only, bounds an expectation, nests token losses inside items, and scores tokens with a generative within-modality likelihood, including image tokens. It does not make the guarantee of Gomes et al. the first of its kind, and it is not a stronger one: the two answer different questions.

Source validation before fusion.. The classical background is outside machine learning. Failure detection and isolation [12, 13], CFAR thresholding [14, 15, 40], and decision fusion with weighted reliabilities [8, 41] are the vocabulary the method borrows. Selecting the set of tokens to keep by looking at their

values makes the missingness informative [42]; Section 3 states the conditions under which withholding a data-selected set is still the right treatment.

3. Method

This section defines the notation, the scores, the calibrated gate, the evidence statistic and seven algorithms. Each algorithm calls the earlier ones, and Algorithm 7 composes all of them. The method has three steps. Each input token is *scored* by its masked log-likelihood inside its own modality with the other modality withheld. Tokens whose score is improbable for clean inputs, at a threshold set on clean data, are *gated*. The decision then *fuses* evidence from the tokens that remain, or from tokens re-infilled by the model. One design rule organises the three steps: evidence that is tested must not be used to gate (Section 3.3).

3.1. Setting and notation

A unified masked discrete diffusion model with absorbing-state masking is given a sequence of caption tokens and image tokens. The network p_θ maps a partly masked sequence to a categorical distribution at every masked position. A pair is $x = (t, v)$ with caption tokens $t = (t_1, \ldots, t_n)$ and image tokens $v = (v_1, \ldots, v_m)$. The modalities are $a, b \in \{T, I\}$ with $a \neq b$. Table 2 lists the symbols. All log-likelihoods are in nats, computed as an `fp32` log-softmax of the logits restricted to the vocabulary of the scored modality.

3.2. Masked scores

For a token j of modality a with observed value x_j, four context levels are distinguished. Let $S_{k(j)}$ be the group of j in a partition Π_K, and let x^a_{-S} denote the tokens of modality a outside S.

$$
\begin{aligned}
A_j &= \log p_\theta\big(x_j \mid x^a_{-S_{k(j)}},\, x^b\big), & B_j &= \log p_\theta\big(x_j \mid x^a_{-S_{k(j)}},\, \varnothing^b\big),\\
C_j &= \log p_\theta\big(x_j \mid \varnothing^a,\, x^b\big), & D_j &= \log p_\theta\big(x_j \mid \varnothing^a,\, \varnothing^b\big). \qquad (1)
\end{aligned}
$$

A_j uses the rest of the own modality and the other modality; B_j uses the own modality only; C_j uses the other modality only; D_j uses nothing and is the position marginal. By subtraction, the total pointwise mutual information

Algorithm 1: MASKEDSCORE: log-likelihood of target tokens under a mask pattern

Input : sequence x; target positions P; extra masked positions X; vocabulary slice $[lo, hi)$ of the targets

Output : $\lambda_j = \log p_\theta(x_j \mid \cdot)$ and $\max_y p_\theta(y \mid \cdot)$ for $j \in P$

Function MASKEDSCORE$(x, P, X, [lo, hi))$**:**

  $\tilde{x} \leftarrow x$; set $\tilde{x}_P \leftarrow \mathsf{M}$ and $\tilde{x}_X \leftarrow \mathsf{M}$;

  $\ell \leftarrow$ logits of $\tilde{x}$; ▷ `one network pass; batched with others`

  for $j \in P$ **do**

    $\pi_j \leftarrow \mathrm{logsoftmax}_{\texttt{fp32}}\big(\ell_{j,[lo,hi)}\big)$;

    $\lambda_j \leftarrow \pi_j[x_j - lo]$ and $\rho_j \leftarrow \exp \max \pi_j$;

  return (λ, ρ);

splits in two ways,

$$A_j - D_j = \underbrace{(B_j - D_j)}_{q_j\text{: intra-modal}} + \underbrace{(A_j - B_j)}_{\text{cross-modal given own context}}$$
$$= \underbrace{(A_j - C_j)}_{\text{intra-modal given } b} + \underbrace{(C_j - D_j)}_{\text{cross-modal}}. \tag{2}$$

The quantity $q_j = B_j - D_j$ asks whether token j is consistent with the rest of its own modality, relative to how likely the value is at that position anyway. It never sees the other modality, and therefore never sees the hypothesis that will be tested. The cross-modal term $C_j - D_j$ is the decision evidence. All four levels come from one primitive, Algorithm 1.

3.3. The gate statistic and the evidence-separation rule

Two regimes are fixed per task before any run. In the regime with *no trusted reference* (detection, verification, matching), neither modality may be used to judge the other, because the cross-modal relation is what is being tested. In the regime with a *trusted reference* (repair with a known clean caption), the trusted modality may be used in the gate, as a predicted state is used in an innovation gate of tracking. The paper studies the first regime.

Evidence-separation rule: in the first regime, the gate may use only statistics computed with the other modality masked, that is B_j and D_j. A gate that used A_j with a candidate caption visible would give each candidate

a different evidence set, and a gate that removed low cross-modal-evidence caption words would remove exactly the words that make a wrong caption wrong. The gate statistic is therefore

$$u_j = -q_j = -(B_j - D_j), \tag{3}$$

where a larger u_j means a more suspicious token. Dividing by the position marginal D_j matters when a corrupted token looks like a typical code: in a model where a corrupted value is drawn from the model's own position marginal independently of context, q_j is the Neyman–Pearson statistic, whereas for a uniform alternative the plain likelihood B_j is (for corruptions that replace tokens by uniform random codes). No proposition about detection power is claimed for corruptions such as JPEG, noise or splice, whose changed codes depend on the content; detection there is an empirical question (Section 4). The score $A_j - B_j$ without the null term cancels the own-context surprise of a corrupted token, because the term appears in both A_j and B_j; it is kept only as a control.

On the text side the gating unit is a word, that is a word-initial piece and its continuation pieces, with the word score the minimum of its piece scores, following the idea that multi-piece words need a word-level score [33]. Gating a word masks all its pieces.

3.4. Calibration by conformal risk control

The threshold is set on a calibration set $\mathcal{D}_{\text{cal}}$ of n_{cal} clean pairs, disjoint by image from every other fold. "Clean" means images as distributed in the data set and passed through the tokenizer; corruption labels are relative to that tokenisation. For item i with m_i gating units and suspicions u_{ij}, the false-alarm proportion at threshold τ is

$$L_i(\tau) = \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{1}\big[u_{ij} > \tau\big] \in [0, 1], \tag{4}$$

which is non-increasing in τ. Define the empirical risk $\widehat{R}(\tau) = n_{\text{cal}}^{-1} \sum_i L_i(\tau)$ and the threshold

$$\hat{\tau} = \inf \Big\{ \tau : \ \frac{n_{\text{cal}}}{n_{\text{cal}} + 1} \, \widehat{R}(\tau) + \frac{1}{n_{\text{cal}} + 1} \leq \alpha \Big\}, \tag{5}$$

with $\hat{\tau} = +\infty$ if the set is empty, so that nothing is flagged. A token is flagged when $u_{ij} > \hat{\tau}$ with a strict inequality.

Algorithm 2: ReliabilityScores: suspicion of every token of one modality

Input : observed pair $\tilde{x}$; modality a; number of groups K; seed
Output : suspicion vector u over the positions of a

Function *ReliabilityScores*$(\tilde{x}, a, K, \text{seed})$**:**
  draw a partition Π_K of the positions of a from the seed;
  for $k = 1, \ldots, K$ **do**
    $B_{S_k} \leftarrow \textsc{MaskedScore}(\tilde{x},\ S_k,\ \text{positions of } b,\ \text{slice of } a)$; ▷ `b withheld`
  $D \leftarrow \textsc{MaskedScore}(\tilde{x},\ \text{all of } a,\ \text{all of } b,\ \text{slice of } a)$; ▷ `layout only; cached`
  $u_j \leftarrow -(B_j - D_j)$ for every position j of a;
  if $a = T$ **then** aggregate u to words by the maximum over the pieces of a word;
  return u;

Proposition 1 (calibration guarantee; conformal risk control applied)**.** *Let the calibration items and a new clean test item be exchangeable as items, and let L_i be defined by Eq.* (4)*, with scores that have no ties. Then $\mathbb{E}\big[L_{\text{test}}(\hat{\tau})\big] \leq \alpha$.*

Proof. The loss is bounded by one and non-increasing in the threshold, so the statement is Theorem 1 of conformal risk control [39] with bound $B = 1$. □

Wording and scope of the guarantee.. The statement made in this paper, and no stronger one, is: the threshold is calibrated so that, for a clean test pair exchangeable with the calibration pairs, the expected proportion of its tokens that are flagged is at most α. Three details keep the statement true in practice. The bound is over items, so tokens inside an item need not be exchangeable or independent and positions may have different score distributions. Ties, which arise from `bf16` logits, are broken by an independent uniform jitter $\epsilon \sim U(0, \epsilon_0)$ added to calibration and test scores, with ϵ_0 below the smallest non-zero gap between distinct calibration scores, computed and not chosen. The statistic, K, α and the fold sizes are fixed before any test item is scored, because choosing them from calibration or test items would break exchangeability. The guarantee is marginal over the draw of the calibration set, so one realised calibration set may lie above α on a finite test set.

Algorithm 3: CalibrateThreshold

Input : calibration set $\mathcal{D}_{\text{cal}}$; modality a; level α; K; seed rule
Data : clean pairs only, disjoint from all other folds
Output : threshold $\hat{\tau}$ and jitter scale ϵ_0, stored with the hash of the item list, K, seed and statistic name

Function *CalibrateThreshold*$(\mathcal{D}_{\text{cal}}, a, \alpha, K)$**:**
  for *each item* i *in* $\mathcal{D}_{\text{cal}}$ **do**
    $u_i \leftarrow \text{ReliabilityScores}(x_i, a, K, \text{seed}_i)$;
  $\epsilon_0 \leftarrow$ a value below the smallest non-zero gap between distinct pooled scores;
  add independent jitter $\epsilon_{ij} \sim U(0, \epsilon_0)$ to every u_{ij};
  evaluate $\widehat{R}(\tau)$ at every distinct pooled value and return the smallest τ satisfying Eq. (5);

What is not guaranteed.. There is no bound on detection power: calibration controls false alarms on clean items only. There is no bound on false alarms among the clean tokens of a *corrupted* item, whose neighbours see contaminated context and can be flagged more often; that rate is measured and reported (Section 4). There is no guarantee under shift between calibration and test, such as another data set, caption style, resolution or tokenizer. Iterating the gate by re-scoring after masking breaks monotonicity in τ, so the bound would not apply to it. Words such as "certified" or "provably reliable" are not used for this gate.

3.5. Gating

The gate applies the frozen threshold to the suspicions of an observed pair. Algorithm 4 computes the gated set G and the reliable set R with the same K, seed rule and statistic as in calibration. No iteration is used.

3.6. Withholding flagged tokens: the marginalise arm

Masking a flagged token tells the network that its value is unknown. For a network that equals the Bayes-optimal denoiser of the absorbing-state objective, masking the set G marginalises it: for a masked target $j \notin G$,

$$p_\theta(x_j \mid x_R,\, x_G = \mathsf{M}) = \sum_{x_G} p(x_j \mid x_R, x_G)\, p(x_G \mid x_R) = p(x_j \mid x_R). \qquad (6)$$

Algorithm 4: GATE (uses Algorithms 2 and 3)

Input : observed pair $\tilde{x}$; modality a; frozen $\hat{\tau}$ and ϵ_0; K
Output : gated set G, reliable set R, suspicion u

Function $\textsc{Gate}(\tilde{x}, a, \hat{\tau}, \epsilon_0, K)$**:**
  $u \leftarrow \textsc{ReliabilityScores}(\tilde{x}, a, K, \text{seed})$;
  add independent jitter $\epsilon_j \sim U(0, \epsilon_0)$ to every u_j;
  $G \leftarrow \{j : u_j > \hat{\tau}\}$; $R \leftarrow$ complement of G;
  return (G, R, u);

This is the standard property of such denoisers [3], not a contribution. A real network approximates it. Structured patterns, such as a contiguous block of masked image tokens together with a fully masked caption, are rare under random per-token masking, so the amortisation error may be larger there; a coherence test on the real model is part of the protocol (Section 4).

When the set G is selected from the observed values, Eq. (6) is not enough. Let Y be the quantity to infer, for example the caption hypothesis. Using $p(Y \mid \tilde{v}_R)$ equals the Bayes answer under the corruption model when four conditions hold: (i) R contains no corrupted token; (ii) corrupted values carry no information about the clean values, which holds for replacement by uniform random codes and fails for JPEG and noise; (iii) corruption locations are independent of the content, which fails where changed codes concentrate in regions in which the tokenizer is unstable; (iv) no clean token is in G, or the clean tokens in G are uninformative about Y. The gate selects on the observed values, so the missingness is informative [42]. Each violation has a cost: residual corrupted evidence in (i), discarded signal in (ii) because a noisy copy would call for a denoising posterior and not for marginalisation, a wrong distribution $p(v_G \mid v_R)$ for selected positions in (iii), and lost clean evidence at rate α in (iv). The prediction stated in advance follows: masking should help most for corruptions that satisfy (ii) and (iii), and least, possibly not at all, for JPEG and noise.

3.7. Image-direction evidence

The decision studied is paired verification: for one image, does the true caption c^+ explain the image better than a wrong caption c^-. The image-direction evidence uses one pass per caption with all image tokens masked and the caption visible. Let $\Lambda_j^c[y]$ be the log-probability of code y at image

position j in that pass, so that

$$\Delta_j = \Lambda_j^{c^+}[\tilde{v}_j] - \Lambda_j^{c^-}[\tilde{v}_j], \qquad \bar{\Delta}_R = \frac{1}{|R|}\sum_{j\in R}\Delta_j, \qquad Y_R = \mathbb{1}\big[\bar{\Delta}_R > 0\big]. \quad (7)$$

The position marginal D_j cancels in the difference for a fixed image, so $\Delta_j = (C_j(c^+) - D_j) - (C_j(c^-) - D_j)$. For a single candidate the per-caption evidence is $E_{I\leftarrow T}(c) = |R|^{-1}\sum_{j\in R}\big(\Lambda_j^c[\tilde{v}_j] - D_j[\tilde{v}_j]\big)$, and $\bar{\Delta}_R = E_{I\leftarrow T}(c^+) - E_{I\leftarrow T}(c^-)$. The logits of this pass do not depend on the image, which enters only through the gathered entries. Hence all arms that differ in the set R (no gate, gate, random gates, a changed-code gate, a denoised copy) reuse the same two passes per item. Ties and an empty set count as wrong, $Y = 0$. Two continuous summaries are used in the secondary analyses,

$$d_R = \frac{\bar{\Delta}_R}{s_R}, \qquad z_R = \sqrt{|R|}\, d_R, \quad (8)$$

where s_R is the sample standard deviation of Δ_j over R. The ratio d_R rises when tokens with extreme per-token evidence are removed for any reason, which is why a random gate and an uncertainty-ranked gate of equal size are reported next to it.

Analysis: when does withholding help an additive statistic?. Consider the statistic $S = \sum_j I_j\Delta_j$ over m tokens, with $I_j \in \{0,1\}$ the indicator that token j is kept. Under an independent-token approximation, clean tokens have mean μ and variance σ^2 and are kept with probability $1-\alpha$, corrupted tokens have mean μ_c and variance σ_c^2 and are kept with probability $1-\beta$, and a fraction ρ of tokens is corrupted. Then

$$\mathbb{E}S = m\big[(1-\rho)(1-\alpha)\mu + \rho(1-\beta)\mu_c\big], \quad (9)$$

$$\mathrm{Var}S = m\Big[(1-\rho)\big\{(1-\alpha)\sigma^2 + \alpha(1-\alpha)\mu^2\big\} + \rho\big\{(1-\beta)\sigma_c^2 + \beta(1-\beta)\mu_c^2\big\}\Big]. \quad (10)$$

The no-gate case is $\alpha = \beta = 0$, and the gate raises $\mathbb{E}S/\sqrt{\mathrm{Var}S}$ if the ratio after exceeds the ratio before. For $\mu_c = 0$ this reduces to

$$\rho\,\sigma_c^2\,(\beta - 2\alpha + \alpha^2) \;>\; \alpha(1-\alpha)(1-\rho)\,(\sigma^2 + \mu^2). \quad (11)$$

Three consequences are stated in advance. A necessary condition is $\beta > 2\alpha - \alpha^2$. A random gate has $\beta = \alpha$, for which the left side is negative, so it

Algorithm 5: ImageEvidence (marginalise arm; uses Algorithms 1 and 4)

Input : observed image tokens $\tilde{v}$; candidate captions $c_1, \ldots, c_N$; reliable set R from Gate or another arm; cached position marginal D

Output : $E_{I \leftarrow T}(c_k)$ for every candidate and, for a pair (c^+, c^-), the paired quantities Y_R, $\bar{\Delta}_R$, d_R

Function *ImageEvidence*$(\tilde{v}, \{c_k\}, R)$**:**

  for *each candidate* c_k **do**

    $\Lambda^{c_k} \leftarrow$ MaskedScore with all image tokens masked and c_k visible, keeping the `fp32` log-probabilities over all image codes at all m positions; ▷ `one pass; reusable by every arm`

    $E_{I \leftarrow T}(c_k) \leftarrow \frac{1}{|R|} \sum_{j \in R} \big(\Lambda_j^{c_k}[\tilde{v}_j] - D_j[\tilde{v}_j]\big)$;

  if $N = 2$ **then** $\Delta_j \leftarrow \Lambda_j^{c^+}[\tilde{v}_j] - \Lambda_j^{c^-}[\tilde{v}_j]$ for $j \in R$; **return** $\big(\mathbb{1}[\bar{\Delta}_R > 0],\ \bar{\Delta}_R,\ \bar{\Delta}_R / s_R\big)$;

  return $\{E_{I \leftarrow T}(c_k)\}$;

lowers the ratio. On a clean item ($\rho = 0$) any gate with $\alpha > 0$ lowers the ratio slightly. The model is an approximation, since neighbouring tokens are correlated, and μ_c need not be zero even for random codes: a caption that makes the predicted image distribution flatter gives random codes a higher probability. The protocol estimates $\rho, \mu, \sigma^2, \mu_c, \sigma_c^2$ per corruption from the per-token terms and compares the predicted sign with the observed one; the model is refuted for a corruption whose observed difference has an interval that excludes zero and whose sign disagrees with the prediction.

3.8. Text-direction evidence and conditioning arms

The text direction scores the caption tokens given the image. With caption c fully masked and the image visible, the evidence is the sum over caption tokens of the log-probability minus the same pass with the image masked,

$$E_{T \leftarrow I}(c) = \sum_j \big(C_j - D_j\big), \tag{12}$$

where the image tokens in the gated set G are masked in the first pass. This is a conditioning arm in which the network, and not the arithmetic of a sum,

withholds the flagged tokens. In the exploratory analyses the statistic is the mean per-token value.

3.9. Re-infilling flagged tokens: the impute arm

Instead of leaving flagged tokens masked, the model can fill them in. Algorithm 6 samples L completions in S_r steps, unmasking the k_s highest-probability positions at step s, and optionally keeps only filled values whose recorded probability reaches κ (hybrid arm). If the filled tokens are generated with the candidate caption visible, the evidence for that caption is biased upward. Writing $\hat{v}_G \sim p(v_G \mid v_R, c)$, Bayes' rule gives

$$\begin{aligned} \mathbb{E}\big[\log p(c \mid v_R, \hat{v}_G)\big] &= \log p(c \mid v_R) + \mathrm{KL}\big(p(v_G \mid v_R, c) \,\|\, p(v_G \mid v_R)\big) \\ &\geq \log p(c \mid v_R), \end{aligned} \tag{13}$$

so every candidate receives a non-negative bonus equal to how strongly it constrains the filled region, which need not favour the true caption. The rule is therefore to re-infill with the caption masked and to run the caption-visible variant only as a leakage control. With exact sampling from $p(v_G \mid v_R)$, the expectation of $p(c \mid v_R, \hat{v}_G)$ equals $p(c \mid v_R)$, so the impute arm is unbiased for the marginalise arm in the probability domain; in the log domain an L-sample estimate is biased downward by Jensen's inequality and adds sampling variance. The preregistered prediction is that marginalise is at least as good as impute on the paired endpoint; a result in which impute is better, with an interval excluding zero, would refute the coherence premise for this model. Re-infilling is part of the method, but it is outside the preregistered main run (Section 4).

3.10. Two-way evidence fusion

For a candidate c there are two directional statistics, $E_{T\leftarrow I}(c)$ of Eq. (12) and the image-direction evidence of Eq. (7). Inverse-variance weighting is not used, because the two statistics estimate different quantities on different scales, share one network and are deterministic for a given input under full masking. The fusion rule is reported in three steps on the paired differences $\Delta^{\text{pair}} = (\Delta_T, \Delta_I)^\top$ with $\Delta_d = E_d(c^+) - E_d(c^-)$: (1) each direction alone; (2) an unfitted Stouffer sum, with each paired margin standardised by its standard deviation on calibration items and the sum divided by $\sqrt{2 + 2\hat{r}}$, where $\hat{r}$ is the calibration correlation; (3) Fisher's linear discriminant,

$$\hat{\omega} = \hat{\Sigma}_\Delta^{-1} \hat{\mu}_\Delta, \qquad F(c) = \hat{\omega}_T \, E_{T\leftarrow I}(c) + \hat{\omega}_I \, E_{I\leftarrow T}(c), \tag{14}$$

Algorithm 6: ReInfill (impute arm; uses Algorithms 1 and 4)

Input : observed image tokens $\tilde{v}$; gated set G; steps S_r; samples L; optional threshold κ

Output : L completed images $\hat{v}^{(\ell)}$ and, for the hybrid arm, a residual gated set G'

Function $\textit{ReInfill}(\tilde{v}, G, S_r, L, \kappa)$**:**

  for $\ell = 1, \ldots, L$ **do**

    $y \leftarrow \tilde{v}$ with $y_G \leftarrow \mathsf{M}$; caption masked ; ▷ `caption masked: no leakage`

    for $s = 1, \ldots, S_r$ **do**

      $U \leftarrow$ positions of G still masked in y;

      $k_s \leftarrow \lceil |U|/(S_r - s + 1) \rceil$;

      $\pi_j \leftarrow p_\theta(\cdot \mid y)$ for $j \in U$; ▷ `one pass, via MaskedScore`

      choose the k_s positions of U with the largest $\max \pi_j$;

      sample $y_j \sim \pi_j$ and record $\pi_j(y_j)$;

    $\hat{v}^{(\ell)} \leftarrow y$;

  if *hybrid* **then** $G' \leftarrow \{j \in G : \pi_j(y_j) < \kappa\}$ and re-mask them;

  return $\{\hat{v}^{(\ell)}\}$ and G';

with the weights fitted on calibration pairs and frozen before the test fold. The fusion therefore has two fitted weights, which is why the paper says "no network finetuning" and not "training-free". The sum of per-token log-likelihood ratios in Eq. (7) is the naive-Bayes rule of distributed detection [8]. Whether a fusion gain can be tested depends on the clean accuracy of the best single direction (Section 4).

3.11. Cost in network calls

Cost is counted in network passes (one row of a batch), because a call at this scale is the dominant term. Table 3 lists the passes per quantity. Gating the image adds K passes per image version and nothing else to the image-direction evidence, because the evidence passes depend on the caption only. With only the image gated, verification of N candidates costs K gate passes, $2N$ evidence passes (the image-direction half cached per caption), and $S_r L$ more for the impute arm, against $2N$ for the ungated two-way baseline.

Algorithm 7: GATEVERIFY: gated cross-modal verification (composes Algorithms 1–6)

Input : observed pair $\tilde{v}$ and candidates $c_1, \ldots, c_N$; frozen $\hat{\tau}_I, \epsilon_0$; fusion weights $\hat{\omega}$; arm $\in$ {none, M, I}

Output : a score $F(c_k)$ per candidate and a decision

Function *GATEVERIFY*$(\tilde{v}, \{c_k\}, \hat{\tau}_I, \hat{\omega}, arm)$**:**

- $(G, R, u) \leftarrow$ GATE$(\tilde{v}, I, \hat{\tau}_I, \epsilon_0, K)$; ▷ `once per image; caption masked, hypothesis-free`
- **if** *arm* $=$ *none* **then** $G \leftarrow \varnothing$; $R \leftarrow \{1, \ldots, m\}$;
- **if** *arm* $= I$ **then**
  - $\hat{v} \leftarrow$ REINFILL$(\tilde{v}, G, S_r, L, \kappa)$; $\tilde{v} \leftarrow \hat{v}$; $R \leftarrow \{1, \ldots, m\}$
- $\{E_{I \leftarrow T}(c_k)\} \leftarrow$ IMAGEEVIDENCE$(\tilde{v}, \{c_k\}, R)$;
- **for** *each candidate* c_k **do**
  - $E_{T \leftarrow I}(c_k) \leftarrow$ MASKEDSCORE on the caption, caption fully masked, image visible with the tokens of G masked (arm M);
  - $F(c_k) \leftarrow \hat{\omega}_T E_{T \leftarrow I}(c_k) + \hat{\omega}_I E_{I \leftarrow T}(c_k)$;
- **return** $\{F(c_k)\}$ and the decision (argmax, or a threshold for unpaired verification);

4. Experimental protocol

The protocol was fixed before any outcome of the main run existed. It is recorded in a preregistration card, a statistical analysis plan with two addenda, and a method specification, all pinned by commit and hash (Section 4.11). This section states what was run, with what, and how the results will be judged. Parts of the method that the preregistered run does not exercise are named as such.

4.1. Model, data and tokenisation

The model is UniDisc with 1.4B parameters [4], used as released, with no training or finetuning here. Images and captions come from the COCO val2017 split [43]. Images are tokenised at 512 px with the VQ-16 tokenizer of LlamaGen [44], giving a 32×32 grid of $m = 1024$ image tokens drawn from 16 384 codes. Captions of 5 to 32 text tokens are used. Instance annotations of val2017 are used to restrict wrong captions (Section 4.2). Whether COCO val2017 images were in the training data of UniDisc or of the tokenizer is

unknown to the authors and is a stated limitation (Section 6). The backbone runs in `bf16`; log-probabilities are an `fp32` log-softmax of the logits restricted to the scored modality. The model is treated as time-independent, as in the released inference path, which calls the network without a noise-level input.

4.2. Folds, negatives and tokeniser check

Four disjoint sets of images are drawn by COCO image id with a fixed seed (20261029): calibration (300 items), headroom (100), test (300, with an ordered extension to 500 used only if the sample-size rule of Section 4.4 asks for it), and the 200 items of an earlier detection pilot, which serve as the detection set. All new folds exclude the 494 image ids of the two pilots. The fold file has SHA-256 `450edb54...f311` (full hash in the data statement). The wrong caption c^{-} of an item is a caption of another image, fixed once per item and shared by every arm and corruption; where annotations exist, it comes from an image that shares no annotated COCO category with the true image. Before any score is computed, the frozen caption token counts are compared with the counts of the runtime tokeniser. An item whose real count falls outside 5 to 32 is excluded from every analysis, counted and not replaced; if more than 10 % of any fold fails the check the run stops and a fold-construction defect is reported.

4.3. Corruptions

Table 4 lists the corruptions. Three are *realistic*: they act on pixels before tokenisation and produce plausible codes. Two are *sanity* corruptions that replace tokens by uniform random codes; they are not in the primary family. The corrupted-token label is $Z_j = \mathbb{1}[\tilde{v}_j \neq v_j^{\circ}]$, relative to the tokenisation of the clean image. Noise draws and the donor image and location of a splice derive from the item id, so one corruption realisation exists per image.

4.4. Headroom rule and sample size

A gate can restore verification only if corruption lowers it and removing the corrupted tokens recovers it. Both are checked on the 100 headroom items, blind to the method: no gate flag is computed on those items. The endpoint is the size-normalised evidence d_R of Eq. (8). For corruption c and rung r let $u_i = d_i^{\text{clean}} - d_i^{\text{cor}}$ and $v_i = d_i^{\text{chg}} - d_i^{\text{cor}}$, where "chg" is the changed-code gate that keeps exactly the tokens that did not change. With T_1 and T_2 the paired t statistics of u and v over the n_v items that have all values, a rung passes if $n_v \geq 90$, $T_1 \geq 2.0$ and $T_2 \geq 2.0$. The first passing rung of a corruption is its

Algorithm 8: GeometryMatchedGate: a random gate with the shape of the gate

Input : gated set G on the 32×32 grid; seed
Output : a random gate G^{geo} of the same size and shape at another place

Function *GeometryMatchedGate*(G, seed)**:**
  draw (d_x, d_y) uniformly from $\{0, \ldots, 31\}^2 \setminus \{(0,0)\}$;
  return $\{((x + d_x) \bmod 32,\ (y + d_y) \bmod 32) : (x, y) \in G\}$;
  ▷ `torus shift`

operating point; a corruption that passes neither rung has none, is not tested, and keeps a p-value of one in the primary family. If no corruption has an operating point, the downstream claim is withdrawn and the paper reports detection, calibration and the clean control only.

The headroom items also fix the test-fold size. Let D_c be the number of headroom items whose paired correctness differs between the changed-code gate and no gate, and $D^* = \max_c D_c$ over corruptions with an operating point. Rule R1 ($D^* \geq 29$): test on the first 300 ids with the correctness endpoint. Rule R2 ($18 \leq D^* < 29$): test on the first 500 ids. Rule R3 ($D^* < 18$): test on the first 300 ids with the paired change of d_R as the primary endpoint and correctness as a secondary one. The decision, D_c and D^* are written to a file that is committed before the first test item is scored.

4.5. Arms and controls

Table 5 lists the arms of the image-direction analysis. All arms reuse the two caption-only passes per item (Table 3). The primary contrast is the marginalise arm (*GATE-M*) against *none*. The controls separate three competing explanations: mask geometry (an equally shaped mask elsewhere), texture or encoder instability (cheap non-generative detectors), and pixel denoising.

4.6. Detection baselines and comparators

Detection of changed codes is evaluated against four non-generative or uncalibrated rivals, all fixed in advance: the *patch texture* (pixel variance of the 16×16 patch of a token), the *neighbour codebook distance* (mean L_2 distance of the token's codebook vector to its eight neighbours), the *code*

rarity (negative log frequency of the code among the calibration fold's clean codes), and the *uncertainty score* s_3 above. Both signs are admitted for each rival, which favours the rival (a sign oracle). Plain-B (B_j without the null term) is the detector of Eq. (3) without the null normalisation, and the score $A_j - B_j$ is the cancellation control (expected near 0.5). The three families of comparator that the related work names are handled as follows. *TbV-score (non-finetuned).* The observed-position probability of Trust but Verify [25] computed from the raw backbone logits of the unmodified model with the caption masked, never from the released output layer, which writes a probability of one at every unmasked position. It reproduces the score of that paper but not its method, which needs a finetune that is outside this study. It is reported threshold-free, and it is marked not applicable if the raw logits turn out to be degenerate. *UniDisc repair recipe* [4]: random mask of equal size, n re-infills, best of n by the model's own likelihood, at matched network calls. This comparator belongs to the re-infill tier, which is not frozen; if it is run, an addendum is written before its first item, and otherwise the paper says that it was not run. *Pixel denoiser*: OpenCV non-local-means denoising for colour images, one setting for all corruptions, with strength $h = h_{\text{color}} = 25.5$ grey levels (matched to the noise level 0.10×255) and windows of 7 and 21 pixels. On 20 images this setting raises the PSNR of the noisy image at $\sigma = 0.10$ from 20.41 to 25.75 dB, and lowers that of JPEG quality 10 from 26.82 to 25.13 dB. The denoiser is therefore not neutral on JPEG: against it GATE-M is compared with an aggressive denoiser that harms the image, and the JPEG row is worded accordingly. Token-Critic [22], CO2Jump [24] and FreeCorrection [23] are discussed but not run: they need a trained critic, are sampler-level methods for generation, or have code that is not runnable at the pinned commit.

4.7. Endpoints and tests

Primary family.. For each of JPEG, noise and splice, at the operating point from the headroom rule, the endpoint is the paired correctness Y of Eq. (7) under GATE-M against none. The test is the exact two-sided McNemar test [45] on the discordant items: with b items right under GATE-M and wrong under none, c items the reverse, and $n_d = b + c$,

$$p = \min\Big\{1,\ 2\,\mathbb{P}\big(\mathrm{Bin}(n_d, \tfrac{1}{2}) \le \min(b, c)\big)\Big\}. \tag{15}$$

The family has exactly three hypotheses at all times, with Holm step-down [46] at familywise level 0.05: a corruption without an operating point stays in

the family with $p = 1$. The headline claim is supported only if at least one hypothesis is rejected after Holm with $b > c$; a rejection with $c > b$ is reported as "GATE-M worse than none". With every test the paper reports b, c, n_d, n, both accuracies, the net difference $(b - c)/n$ with paired bootstrap intervals (10 000 resamples) at 95 % and at 98.33 %, and the Clopper–Pearson interval [47] of b/n_d. The design minimum detectable net gain at $n = 300$ is 0.053 unadjusted and 0.060 at the first Holm step, under an assumed discordance of 10 % (assumed, not observed).

Secondary families (each with its own Holm correction, none able to rescue a failed primary test).. Table 6 lists them. Everything else is exploratory, reported with intervals and without p-values that enter a claim.

Calibration diagnostics.. On the clean test items, the realised false-alarm proportion $\hat{F} = n^{-1} \sum_i L_i(\hat{\tau})$ is reported for $\alpha \in \{0.01, 0.05, 0.10\}$, with $\alpha = 0.05$ primary and the others unselected sensitivities. A violation is flagged when $Z_{\text{cal}} = (\hat{F} - \alpha)/\left(\hat{\sigma}_L \sqrt{1/n_{\text{cal}} + 1/n_{\text{test}}}\right)$ exceeds 2.13 at any level, where $\hat{\sigma}_L$ is the pooled item-level standard deviation of L_i. A pass means "no evidence of violation" and is never worded as validation. Reported with equal prominence: the precision among flagged tokens per corruption, the false-alarm proportion on the clean tokens of corrupted items, and the spatial map of clean false alarms. For $\alpha = 0.05$ on an 8×8 patch image, about 48 false flags are expected against at most 64 true ones, so precision is at most 0.57 even for a perfect detector.

Pipeline checks and positive controls.. For the sanity corruptions, the lower bound of the 95 % interval of the per-item AUROC of q and of plain-B against the changed-code labels must exceed 0.5; a result at chance is a pipeline defect, not a scientific result, and triggers an audit without any change of rule. The clean no-gate accuracy of the image direction is reported with a Wilson interval against the pilot value of Section 4.9 as a descriptive pipeline check; it is not a validation and tests nothing. A coherence test measures, on 100 clean calibration items, whether the denoiser's joint likelihood of two image tokens is independent of the order in which they are revealed. Its pre-stated reading is "coherent within numerical noise" if the median absolute gap lies within the 95th percentile of the gap between repeats of the same computation, and "incoherent at the evidence scale" if it exceeds the clean mean per-token paired evidence; in the second case Eq. (6) and the marginalise–impute comparison are reported as not applicable at the scale that matters.

Reading of results.. The outcome is labelled *positive* if at least one primary hypothesis is rejected after Holm with $b > c$, which supports only "GATE-M improves image-direction verification over no gate on that corruption"; attribution to detection needs the mechanism family. It is *negative* if no primary hypothesis is rejected and, for every corruption with an operating point, the 90 % paired bootstrap interval of the net accuracy difference lies inside $(-0.05, +0.05)$ with at least 240 completed items. It is *inconclusive* otherwise. A null result is worded "no improvement was detected; the data are compatible with a net accuracy change between L and U", and never "GATE-M has no effect". Under rule R3 the primary endpoint is the evidence ratio, the label negative is not available, and the positive sentence says that GATE-M increased the evidence ratio and nothing about verification accuracy, with the caveat that removing heavy-tailed tokens can raise a ratio for any reason. Rules that stop the claim are fixed in advance: no corruption with an operating point, detection by q not better than the best rival on both JPEG and noise, no primary rejection, a Z_{cal} above 2.13, and more than 5 % empty or non-finite evidence in an arm.

4.8. Two-way fusion and the text direction

Two-way fusion is analysed only if the best single direction has a clean test accuracy below 0.95. Otherwise the paper states that the fusion gain cannot be tested at that ceiling. The text-direction conditioning arms (none, GATE-M with the flagged image tokens masked, geometry-matched, changed-code) run for the three realistic corruptions and are exploratory. Caption corruption, re-infilling and a hard-negative set are not part of the preregistered main run. The title, abstract and highlights therefore make no claim about fusion, marginalisation versus imputation, re-infilling or caption corruption, and the paper states that these were not tested unless an addendum adds them before the first item of the run concerned.

4.9. Exploratory feasibility result

A pilot on 300 clean COCO images, run before the protocol was fixed, shows the pair-level signal on which the endpoint rests. For each image the true caption was compared with a caption drawn from a different image, and accuracy is the share of images for which the true caption scores higher (chance 0.5; bootstrap intervals over images, 10 000 resamples). With $K = 8$ complementary masks and the other modality visible, the text direction reaches 0.730 (95 % interval 0.680 to 0.777) and the image direction 0.783

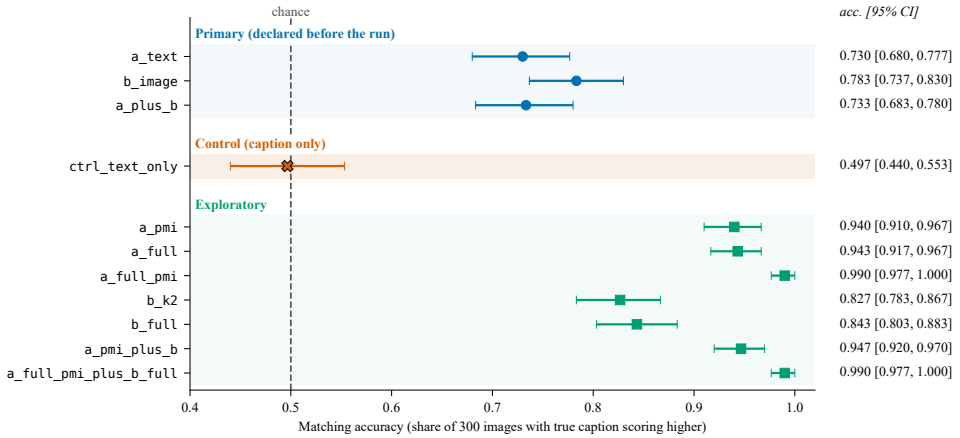


Figure 1: Matching accuracy of UniDisc 1.4B on true versus mismatched COCO captions (exploratory feasibility result, 300 images). Markers show the share of images for which the true caption scores higher (chance 0.5, dashed); bars are 95 % bootstrap intervals over images (10 000 resamples). Blue: primary scores, text direction a_text and image direction b_image with $K = 8$ complementary masks, and their sum. Orange: caption-only control, which stays at chance. Green: variants chosen after the primary results (image-masked null subtracted, single full-mask pass, $K = 2$, sums). One run, one mismatch draw; the intervals do not cover variation across draws or models.

(0.737 to 0.830). A caption-only control, with the whole image masked, stays at chance, 0.497 (0.440 to 0.553), so the text-direction value is not explained by caption fluency. Variants chosen after seeing these results, with all tokens of the scored modality masked in one pass and the image-masked null subtracted, reach 0.990 (0.977 to 1.000) in the text direction and 0.843 (0.803 to 0.883) in the image direction, 253 of 300 images (Figure 1). The image-direction accuracy of 0.843 is the headroom that makes it, and not the text direction, the primary endpoint: the text direction sits near a ceiling at which restoration cannot be shown. The mean paired margin of that image-direction statistic is 0.18 nats per token. This is a single run with one mismatch draw, on clean images, with the full-mask variants selected post hoc; it is exploratory and carries no inference beyond the intervals shown.

4.10. Hardware, runtime and missing items

All runs use one laptop GPU (NVIDIA RTX 5070 Ti Laptop, 12 GB) with 31 GB of system memory and no other hardware. In the pilot, one item cost 58 network passes, the median item time was 7.0 s, and the peak GPU memory was 8.2 GB, which is why passes are batched four rows at a time.

Runs go in short checkpointed chunks under a guard that stops on memory, temperature and power-source events and is resumable; an abort ends the stage and is not retried until reviewed. An earlier detection pilot stopped after 20 complete items when the GPU session failed. Missing items are handled by a fixed rule: complete cases per analysis, no replacements and no reordering, so that a crash leaves a prefix of the item list; if the completed base fold has fewer than 240 items, the paper says that the fold is under 80 % complete, prints the minimum detectable effect at the achieved n, and does not read "not rejected" as evidence against GATE-M. No analysis that computes Y, b, c, p or a difference of d_R is run on test data before the planned fold is complete. Mask-seed and corruption-seed variance are not in the intervals, which is a stated limitation.

4.11. Preregistration, pins and disclosed deviations

The code, the tests, the fold file and the analysis plan are pinned: the code at commit `164a18a5`, the analysis plan by SHA-256 (`b4e0bf57...6c47`), and the denoiser check, crash diagnosis and audit files by hash in the repository. Changes after the pin are logged as amendments; none is made inside a stage. One analysis preceded the analysis plan and is disclosed. The first 21 items of the detection set were analysed on CPU before the plan's second addendum, without the gate statistic q (which needs a GPU pass). In that exploratory look the mean per-item AUROC against changed-code labels of plain-B was 0.54 on JPEG quality 10 and 0.51 on noise 0.10, while pixel variance with its sign flipped scored 0.76 and 0.77. The sign of the pixel-variance rival therefore cannot be claimed to be fixed blind, which is why both signs of every rival are admitted and the rival is chosen on the data. Any change to the score, K, α, folds, rungs, arms, family or rival rule that was made after that look and is not a new version of the preregistration is listed in the change log of the preregistration card and disclosed here. **[AUTHOR: final list of amendments and the date of the first M2 outcome; confirm the status of the open decisions on guard thresholds, the JPEG denoiser setting and the fallback before the run.]**

5. Results

[RESULT: this section is filled only from the frozen run directories after the planned folds are complete. Lead each subsection with

the observation, then the interval, then the preregistered wording from Section 4.7. Print the metric direction in every table header.]

5.1. Headroom and operating points

[RESULT (run P2-GATE-M3): table of the headroom check, per corruption and rung: n_v, T_1, T_2, pass or fail, the chosen operating point, D_c, D^* and the rule (R1, R2 or R3) with the test-fold size. State which corruptions have no operating point and what that means for the primary family. If no corruption has an operating point, say that the downstream claim is withdrawn.]

5.2. Calibration of the gate

[RESULT (runs P2-GATE-M2, P2-GATE-M4 clean arm): realised false-alarm proportion $\hat{F}$ with item-clustered interval for $\alpha = 0.01, 0.05, 0.10$; Z_{cal} and the verdict ("no evidence of violation" or violation); precision among flagged tokens per corruption; false-alarm proportion on the clean tokens of corrupted items; spatial map of clean false alarms (figure); the exploratory 2 000-split check of the marginal statement.]

5.3. Primary result: GATE-M against no gate

[RESULT (run P2-GATE-M4): complete Table 7; the sentence that applies from the preregistered wording table (positive, negative or inconclusive); the three-way label; n_d and, for a null, the statement "only $\theta \geq \theta^*$ would have been detectable" (a sensitivity statement, not an observed power).]

5.4. Mechanism controls

[RESULT: GATE-M against the geometry-matched random gate (seed 1 confirmatory; seeds 2 and 3 as replications), the scattered random gate, plain-B top-k and s_3 top-k (counts b, c beside the primary result, especially under rule R3), the changed-code gate, and GATE on clean images. Use the wording of the mechanism rule: if the mechanism test is not rejected, the improvement is "not distinguishable from masking the same shape elsewhere".]

5.5. Detection against cheap rivals

[RESULT (runs P2-PROBE-02 resumed, P2-GATE-M4): per-item AUROC of q, plain-B, s_3, $A_j - B_j$, texture, neighbour distance, rarity and TbV-score (non-finetuned) against changed-code labels, per corruption, with the best cheap rival, its sign, and the paired difference with its interval; items excluded for too few corrupted or clean tokens, counted. State in the same sentence that the rival is chosen on the data. Sanity corruptions (patch, swap): the chance-bar check.]

5.6. Pixel denoising as a defence

[RESULT: GATE-M against the denoised copy on JPEG and noise; for JPEG use the wording that the denoiser harms the image at this setting.]

5.7. Evidence ratio and the independent-token model

[RESULT: paired change of d_R (and z_R as sensitivity) per corruption; the plug-in estimates of $\rho, \mu, \sigma^2, \mu_c, \sigma_c^2, \alpha_{\text{obs}}, \beta_{\text{obs}}$ from the per-token terms, the predicted sign of Eq. (11)'s ratio change next to the observed sign, and whether the pre-stated refutation applies.]

5.8. Coherence of the denoiser across mask patterns

[RESULT: the median absolute gap against the 95th percentile of the repeat gap, and the pre-stated reading.]

5.9. Text-direction conditioning arms and two-way fusion (exploratory)

[RESULT: text-direction arms for the three realistic corruptions with intervals. Fusion only if the best single direction has a clean test accuracy below 0.95; otherwise one sentence that the fusion gain cannot be tested at the ceiling, with the clean accuracy.]

5.10. Cost

[RESULT: network passes per item by stage, median time per item, peak GPU memory, from the run configuration and telemetry files.]

6. Discussion and limitations

6.1. Findings

[RESULT: for each finding, the observation, the likely cause marked as a hypothesis, and what would falsify it (style rule 26). Use the three-way label of Section 4.7; say "no improvement was detected" for a null. Do not describe the gate as improving verification, repair or generation unless a primary rejection with $b > c$ exists.]

6.2. Where the work sits in information fusion

[AUTHOR: after the results are frozen, state in two or three sentences what is fused (image tokens as sources, an additive log-likelihood-ratio evidence sum), what the gate adds as source validation, and what is not claimed (a new fusion rule). Settle the scope-fit question with the supervising professor before submission; the open check is to give the abstract and Figure 1 caption to a reader with no project context and ask what is fused.]

6.3. Limitations

The limitations below are fixed by the design and do not depend on the outcome.

- *One model, one data set.* All measurements use UniDisc 1.4B and COCO val2017 images as tokenised by one tokenizer. No statement is made for other models, resolutions, caption styles or data sets.

- *Contamination is unknown.* Whether the evaluation images were in the training data of the model or of the tokenizer was not established, and it could make clean pairs easier than unseen ones.

- *Narrow guarantee.* The calibration bounds the expected false-alarm proportion on clean, exchangeable test pairs. It gives no bound on detection power, none on the clean tokens of corrupted items, and none under shift. The realised proportion on one finite test set can lie above α because the statement is marginal over the calibration draw.

- *Corruptions and labels.* Corruption labels are relative to the tokenisation of the clean image, so a "clean" token may already carry compression artefacts of the original photograph. Changed codes under JPEG and noise concentrate where the tokenizer is unstable, and the changed-code gate is not an upper bound for them.

- *Comparators chosen on the data.* The best cheap rival is selected among about nine candidates, with both signs admitted, which biases the detection contrast against the gate. The rival's sign could not be fixed blind (Section 4.11).

- *Pixel denoiser.* One setting was used for all corruptions; it was chosen after one exploratory evaluation on 20 images and harms JPEG images, so the denoising contrast on JPEG favours the gate.

- *Variance not in the intervals.* Intervals cover item sampling. They do not cover variation across mask partitions, corruption draws, jitter seeds or models, because no training seeds exist and only one corruption draw is made per image.

- *Scope of the main run.* The preregistered run exercises withholding by dropping terms of a sum computed from a caption-only pass. Re-infilling, caption corruption, hard negatives, the repair recipe of the backbone's authors, and a second model were not run unless an addendum says so; two-way fusion is testable only below the stated ceiling.

- *Hardware and missing data.* The GPU is a single laptop device that failed in earlier pilots. A crash leaves a prefix of the item list and the missing-item rule applies; a fold under 240 items is reported as incomplete.

- *Narrow novelty.* Detecting and remasking corrupted input tokens, cross-modal remasking, risk-controlled thresholds for vision-language token errors and masked-likelihood scores each have precedents (Section 2). The claim is a measurement of their composition on one model, and it holds only as far as the measured results go.

- *Concurrent work.* A search for concurrent work was last run on 28 October 2026; unpublished or under-review work is not visible. **[AUTHOR: repeat the search once before submission.]**

7. Conclusion

[RESULT, sentence 1: the question and the measurement, stated as in the contribution paragraph, scoped to one pretrained model and one data set.]

[RESULT, sentence 2: the headroom outcome and the primary contrast per corruption, with the number, interval and preregistered wording.]

[RESULT, sentence 3: the calibration result (realised false-alarm proportion and the verdict) and the mechanism and detection controls.]

[RESULT, sentence 4: limits in one sentence (one model, one data set, narrow guarantee, no re-infill or caption corruption in the main run).]

[AUTHOR, sentence 5: future work in one sentence (a second backbone; the re-infill and caption-corruption tiers; a finetuned variant of the observed-position score).]

CRediT authorship contribution statement

Shlok Kumar Goenka: [AUTHOR: choose from the 14 CRediT roles (conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing – original draft, writing – review and editing, visualization, ...)]. [AUTHOR: supervising professor]: [AUTHOR: roles (for example supervision, project administration, writing – review and editing)].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[AUTHOR (Shlok): exact wording of this declaration. The Elsevier template sentence is: "During the preparation of this work the author(s) used [NAME OF TOOL/SERVICE] to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article." Name each tool, its version and its purpose, and the extent of review. Decision still open for Shlok: whether any AI-assisted step of the research process (code, analysis) must

also appear in the methods under the journal's policy; the project rule keeps every AI mention in this section.]

Declaration of competing interest

[AUTHOR: complete the Elsevier declarations tool and state "The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper" only if true.]

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Data availability

[AUTHOR: repository and DOI. Planned content: the code at commit `164a18a5` (`exp_p2/`, `tests/exp_p2/`), the fold file (SHA-256 `450edb543779de7ee5e6a52422fd6507ef52f15ba3b956a8b05c2e561620f311`), the preregistration card, the analysis plan (SHA-256 `b4e0bf575dfc2524c0f9d3d5c6b5f52e3185088a94ced8c75f092104ed396c47`), per-item scores and predictions of every run directory cited, the analysis and figure scripts, and the licence. The COCO images and captions and the UniDisc and tokenizer weights are third-party and are cited, not redistributed. Do not write "available on request".]

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Table 1: Closest works on the four axes of Section 2. Entries restate what the cited papers report, at the level of detail read for this study; "–" means not applicable. AR: autoregressive. FAP: false-alarm proportion.

Work	Tokens treated as suspect	Score	Threshold	Use of flagged tokens
Trust but Verify [25]	Corrupted context tokens, text only	Probability of the observed token at its unmasked position (finetuned 170M model)	Hand-annealed, edit budget from an assumed rate; no guarantee	Re-noise to mask, refill
Gomes et al. [26]	Caption words given an image	Change of CLIP-Score under masking	Conformal risk control, high-probability bound, foil-labelled calibration	Flag only
CO2Jump [24]	Tokens generated by the sampler	Self-confidence and a cross-modal confidence term	Per-step quota from the schedule	Remask, resample
FreeCorrection [23]	Tokens committed by the sampler	Likelihood of the committed token with it remasked	Lowest scores per step, schedule tuned on a validation set	Remask, resample
UniDisc, repair demo [4]	Random positions	–	–	Random mask, reunmask, best of n by model likelihood
Dropout decoding [30]	Visual tokens in an AR vision-language model	Divergence of a logit-lens distribution from the image mean	Hand-set drop probabilities	Drop in several masks, vote
This paper	Input image tokens of a given pair	Masked likelihood in own modality, other modality withheld, normalised by the position marginal	Conformal risk control on clean pairs, expected FAP at most α	Withhold (marginalise) or re-infill

Table 2: Notation.

Symbol	Meaning
$x = (t, v)$	one pair of caption tokens t (n tokens) and image tokens v (m tokens)
$\tilde{x} = (\tilde{t}, \tilde{v})$	the observed, possibly corrupted pair; v° the tokens of the clean image
$Z_j = \mathbb{1}[\tilde{v}_j \neq v_j^\circ]$	corrupted-token label (used only for evaluation)
M	the mask token (M means "unknown")
a, b	scored modality and the other modality
$\varnothing^a$	all tokens of modality a masked
$p_\theta(x_j = \cdot \mid \cdot)$	categorical of the network at masked position j
Π_K = $\{S_1, \ldots, S_K\}$	random partition of the positions of modality a into K complementary groups; $k(j)$ is the group of j
A_j, B_j, C_j, D_j	four context levels of Eq. (1)
q_j, u_j	hypothesis-free intra-modal evidence $B_j - D_j$ and suspicion $u_j = -q_j$
α, $\hat{\tau}$	target false-alarm level and calibrated threshold
G, R	gated (flagged) set and reliable set, R the complement of G
c, c^+, c^-	a candidate caption, the true caption and a wrong caption
$\Lambda_j^c[y]$	image-direction log-probability of code y at image position j with the image fully masked and c visible
Δ_j, d_R, z_R	paired per-token evidence, size-normalised evidence ratio and evidence z-score over a set R
S_r, L	number of re-infill steps and of re-infill samples

Table 3: Network passes per quantity. K: complementary groups; N: candidate captions; S_r, L: re-infill steps and samples.

Quantity	Passes	Depends on, and cache
B over all image tokens	K	image only (caption masked); per image
C for image tokens (image fully masked)	1	caption only; per caption, shared across images and arms
D for image or caption tokens	1	layout only (caption length); per length
A over all image tokens	K	image and caption
Image-direction evidence, all arms	2	the two captions; arms differ only in gathered entries
Text-direction evidence per candidate and arm	1	image, gated set
Re-infill (impute arm)	$S_r L$	image, gated set; shared by all candidates

Table 4: Corruptions. Realistic corruptions are walked in order by the headroom rule (Section 4.4); the first rung that passes is the operating point. Sanity corruptions have one setting.

Corruption	Rungs	Definition
JPEG (realistic)	quality 10, then 5	JPEG compression of the 512-px pixels, then tokenisation
Gaussian noise (realistic)	$\sigma = 0.10$, then 0.20	noise on pixels scaled to $[0, 1]$, clipped and rounded to 8 bit, then tokenisation
Pixel splice (realistic)	128×128, then 192×192 px	a region from a donor COCO image copied to a uniformly random location
Patch (sanity)	one 8×8 block	a block of the token grid replaced by uniform random codes
Swap (sanity)	15 % of tokens	tokens replaced by uniform random codes

Table 5: Arms of the image-direction analysis. R is the set of image tokens kept in the sum of Eq. (7).

Arm	Kept set R	Role
none	all 1024 tokens of the corrupted image	reference
GATE-M	tokens not flagged by the frozen gate (Algorithm 4)	primary arm
geometry-matched random gate	complement of Algorithm 8 applied to GATE-M's mask, 3 seeds	mechanism control
scattered random gate	complement of equally many random tokens, 3 seeds	mechanism control
changed-code gate	tokens whose code did not change	upper reference for sanity corruptions only
plain-B top-k	complement of the $\lvert G\rvert$ tokens with the lowest B_j	gate without calibration, equal size
s_3 top-k	complement of the $\lvert G\rvert$ tokens with the largest $1 - \max_y p_\theta$	uncertainty-ranked gate, equal size
pixel denoise	none, on a denoised copy of the image (JPEG and noise)	defence comparator
GATE on clean images	as GATE-M on uncorrupted images	negative control

Table 6: Secondary confirmatory families. m: number of hypotheses in the family's Holm correction.

Family	Contrast and test	m
Mechanism	GATE-M versus the geometry-matched random gate (seed 1), exact McNemar; tested only for corruptions rejected in the primary family	3
Evidence ratio	paired change of d_R, GATE-M minus none, Wilcoxon signed rank	3
Denoising	GATE-M versus the pixel denoiser, exact McNemar; JPEG and noise only	2
Detection	per-item AUROC of q minus that of the best cheap rival on the 200 detection items, JPEG quality 10 and noise 0.10; paired bootstrap over items	2
Out-of-sample detection	the same contrast with the rival frozen after the detection run, on the test items at their operating points	2
Clean control	GATE-M versus none on clean test images, exact McNemar	1

Table 7: Primary family: GATE-M versus no gate, paired correctness of the image-direction statistic (higher is better). **[RESULT: test fold size, rule, interval type.]**

Corruption (rung)	n	Acc. none	Acc. GATE-M	b / c	$(b-c)/n$ [95 %]	p	p_{Holm}
JPEG	[R]	[R]	[R]	[R]	[R]	[R]	[R]
Gaussian noise	[R]	[R]	[R]	[R]	[R]	[R]	[R]
Pixel splice	[R]	[R]	[R]	[R]	[R]	[R]	[R]